

HADI ALKHUB

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SUMMARY

Software engineering (Honours) student at Griffith University, focused on low-level programming, computer science theory, and sports engineering. I write systems close to the hardware — Linux drivers, bare-metal VGA, ARM64 assembly, a Git object store in Rust — and train competitive programming. Looking for a first research or engineering internship.

EDUCATION

Griffith University

Brisbane, QLD

Bachelor of Engineering, Software Engineering (Honours)

Jul. 2024 – Nov. 2027 (expected)

- GPA 6.3/7.0. Degree focus: low-level programming.
- Coursework: Advanced Computer Systems, Embedded Systems, Full Stack Development.
- Prior BIT year at Griffith (Jul. 2023 – Jun. 2024), then transferred into Software Engineering.

University of Jordan

Amman, Jordan

Coursework in German and English Literature

Oct. 2022 – Jun. 2023

- GPA 3.91/4.00. Enrolled to take selected subjects of interest (not a completed degree).

SELECTED PROJECTS

MAX7219 Device Driver & Digital Clock | *C, Linux kernel, Make*

2026

- Linux character driver (`/dev/max_screen`) for an 8-digit MAX7219 on a Raspberry Pi 4: bit-banged SPI, `ioctl` brightness, and a user-space clock with date/time, flashing seconds, and a scrolling alarm.

VGA Pong on Tiva TM4C129 | *Bare-metal C, Keil*

2025

- Software VGA on a Tiva TM4C1294NCPDT: timer HSYNC/VSYNC, GPIO RGB, a 20×30 framebuffer stretched to a full monitor (SRAM limit), ADC paddle input, and a simple opponent.

hgit | *Rust*

2026

- Git-style object database in Rust: `zlib` + SHA-1 blobs/trees/commits, with `init`, `hash-object`, `cat-file`, `write-tree`, and `commit-tree`.

Chat App System | *MEAN (MongoDB, Express, Angular, Node.js), Socket.io*

2026

- Real-time group chat: Super Admin / Group Admin / user roles, rooms, bcrypt auth, Socket.io messaging, and last-5-message retention in MongoDB.

Algorithm Visualizer | *TypeScript, React, Vite, Tailwind*

2026

- Interactive DSA visualizer (sorting, *d*-ary heaps, AVL trees, prefix/postfix evaluation).

ARM64 Assembly Calculator | *ARM64 assembly*

2026

- Command-line integer calculator in raw ARM64 assembly for Apple Silicon: ASCII conversion, `+`, `-`, `×`, `÷`, and macOS syscalls (no libc arithmetic).

COMPETITIVE PROGRAMMING

Kattis (handle `hadialk4`): ranked **24th in Australia** (score 115.9).

ENGINEERING WORKFLOW

While I prefer writing the code myself, I understand that when the work is specified to use agents so progress is not capped at typing speed. Daily driver: Cursor — Herder CLI, reusable skills, and cloud agents in isolated git worktrees, with design calls staying with me. Currently testing Grok Bot to start those cloud agents, watch the run, and follow up without babysitting a laptop. I trial new tools as they appear and keep only what survives real work.

TECHNICAL SKILLS

Languages: C, C++, Rust, ARM64 assembly, Python, TypeScript, JavaScript, C#

Systems & tools: Linux kernel modules, OpenGL/GLUT, React, Vite, Tailwind CSS, Astro, Keil uVision, Raspberry Pi, Tiva TM4C129, Node.js, Express, Angular, MongoDB, Cursor, Grok Bot

Spoken: English, Arabic, German (A2)